

“Second Project – last – Forecasting in Python”

“Course: Global Supply Chain Analytics”

“Group Members: Nikhith Reddy Pasham, Manaswini Gopala”

Data:

“1. Context & Purpose”

The project aims to prepare and forecast sales data to help managers make better business decisions. The dataset contains several years of retail sales records across product categories and subcategories. The goal is to clean and organize the data, then build forecasts to understand future sales trends.

Good data preparation is critical because poor-quality data leads to poor predictions, “*garbage in, garbage out.*” The data was aggregated monthly for each subcategory to make patterns clearer and forecasts more stable.

The historical horizon (2014–2017) provides sufficient data for trend and seasonality detection. Data were aggregated monthly at the subcategory level to balance temporal granularity with signal stability. The analysis uses total profit to identify the five most profitable subcategories ('Phones', 'Chairs', 'Storage', 'Binders', 'Machines'). This introduces an acknowledged limitation: subcategories with high sales but low margins may dominate the ranking, even if they are less efficient at generating profit.

Main data risks include missing months, sudden sales spikes from promotions, and low-activity categories with irregular data. These were handled by focusing on consistent time periods and choosing subcategories with steady sales history for accurate forecasting.

“2. Method Overview (Python Workflow)”

The project followed a full Python-based workflow from raw data to model comparison:

“Loading & Cleaning:” The raw Superstore dataset was read from Excel. Dates were fixed from Excel serials, numerical fields coerced to float, and duplicates removed using business keys (Order ID, Product ID, Order Date). Missing or invalid entries were dropped to maintain integrity.

“Aggregation & Preparation:” Transactions were grouped by **subcategory and month**, computing both **total sales and total profit**. The time series was resampled to ensure a consistent **monthly frequency**.

“Outlier & Missing Value Handling:” Zero-profit or zero-sales months were flagged and examined. Winsorization was applied, and outliers are consistent. Sparse or incomplete subcategories were excluded from modeling.

“Stationarity & Seasonality Checks:” Trend and seasonality were assessed using rolling statistics, STL decomposition, and ACF. A 12-month seasonal cycle was confirmed for the majority of subcategories; models were fit with seasonality=12 where diagnostics supported it, otherwise non-seasonal specifications were allowed via model selection.

“Modeling Flow:” For each of the five most profitable subcategories, three forecasting methods were built and compared:

- **“Baseline:”** Seasonal Naïve- repeats the value from the same month in the previous year.
- **“Model A:”** Holt-Winters (Exponential Smoothing) with additive trend and 12-month seasonality.
- **“Model B:”** XGBoost regression model using lagged sales, rolling statistics, and calendar features (month, year, seasonal sine/cosine).

“Holdout Strategy:” Time-based splits were applied within 2014–2017 using **rolling train-test windows** for backtesting. Each window trained on earlier months and validated on the following 6–12 months to mimic real forecasting conditions. Final 2018 forecasts were then generated using the best-performing model.

“Assumptions”

- Monthly resampling and 12-month seasonality.
- Consistent subcategory-level aggregation.
- Exclusion of irregular or incomplete series to ensure fair comparison.

“3. Forecasting Focus”

The analysis focused on the five most profitable subcategories, selected based on total historical sales and profit performance. These categories showed consistent monthly data across the 2014–2017 period, making them suitable for reliable forecasting.

Rank	Subcategory	Total Profit (USD)
1	Phones	427,270.44
2	Chairs	365,412.70
3	Storage	331,206.12
4	Binders	302,053.96
5	Machines	239,108.19

“Table 1: Top Five Most Profitable Subcategories (2014–2017)”

“Justification for Selection:” These subcategories were chosen because they consistently generated high revenue and profit across the observed years. Their sales patterns were dense and regular, with minimal missing months, which improves model accuracy.

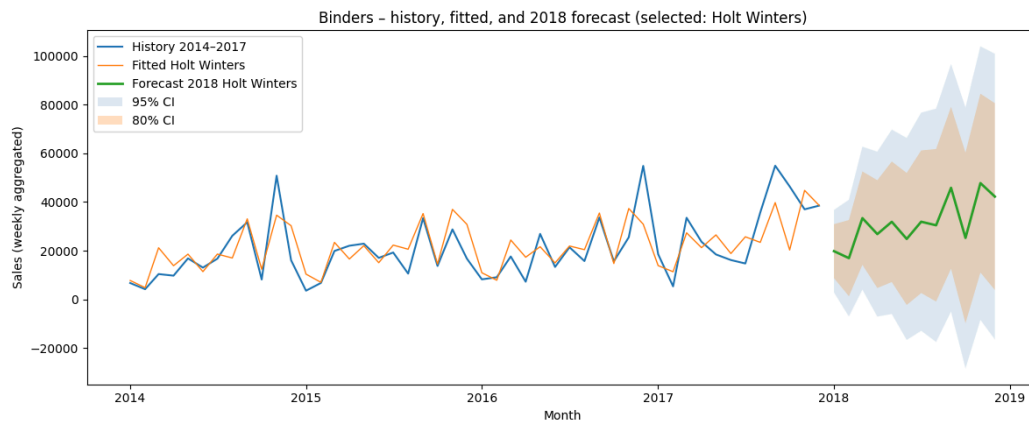
“Sensitivity Note:” Profit and sales rankings were broadly aligned in this dataset. However, profit margins could shift rankings in some cases, subcategories with high sales but low margins might appear

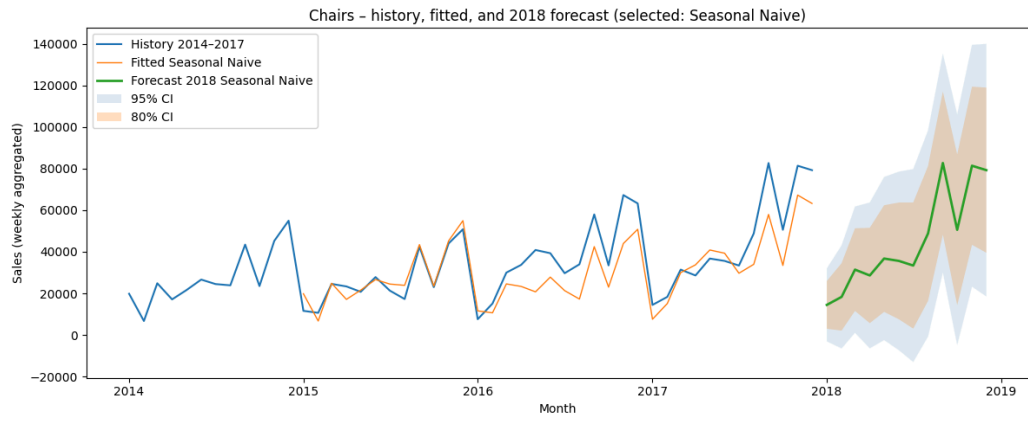
more profitable than they truly are. Here, sales and profit trends moved together, so both metrics supported the same top-five selection.

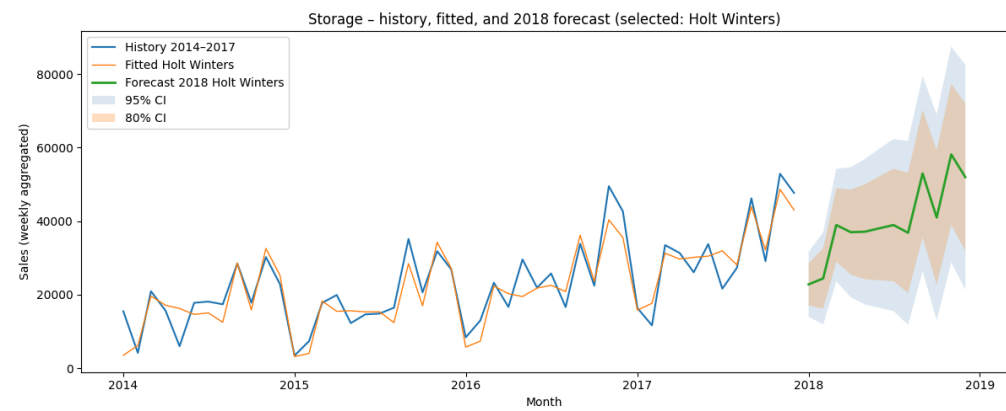
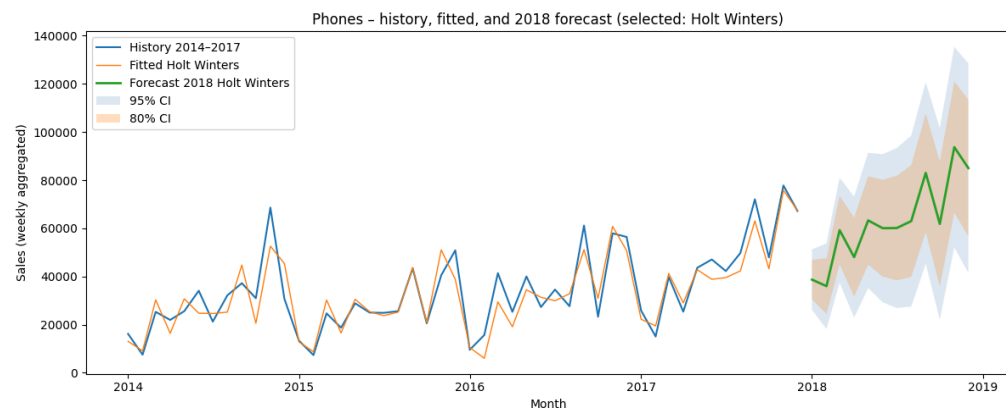
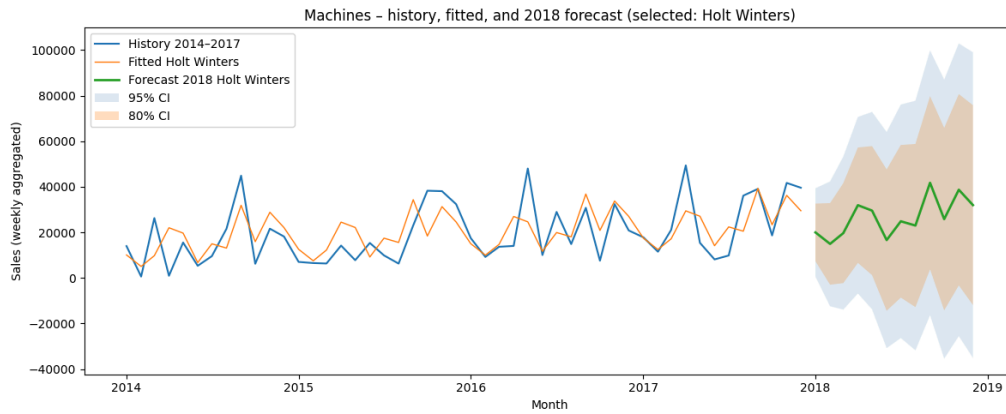
“Forecasting Approaches:” Each subcategory was forecasted using three models:

- **“Baseline:”** Seasonal Naïve model using last year’s same-month value.
- **“Model A:”** Holt-Winters Exponential Smoothing capturing level, trend, and seasonal components.
- **“Model B:”** XGBoost regression with lagged sales and calendar-based features.

“Rationale for Model Choice:” The Seasonal Naïve model served as a benchmark for minimal forecasting skill. Holt-Winters provided an interpretable, well-established approach for seasonal retail data. XGBoost, while more complex, offered flexibility to capture nonlinear effects and subtle interactions. Together, they balance interpretability, scalability, and predictive power for managerial use.







“4. Results & Insights”

The table below summarizes model performance across the top five subcategories using the three forecasting methods. Metrics were averaged over test periods in 2016–2017, with 2018 serving as the final forecast horizon.

Subcategory	Selected Model	MAE	RMSE	MAPE (%)
Binders	Holt–Winters	9,324.22	10,365.21	44.89
Chairs	Seasonal Naïve	9,494.80	10,530.42	25.69
Machines	Holt–Winters	9,298.52	12,741.52	41.09
Phones	Holt–Winters	7,488.54	8,586.88	22.17
Storage	Holt–Winters	5,104.24	6,684.79	23.37

“Table 2: Model Performance by Subcategory (Average Test Metrics)”

“Key Findings”

- **“Phones”** continued to show a strong upward trend with consistent seasonal peaks, reflecting sustained market demand and effective promotional cycles. Forecasts for 2018 indicate further growth with widening confidence intervals toward year-end.
- **“Chairs”** maintained a steady sales trajectory, with stable repeating patterns across years. The Seasonal Naïve model captured this well, producing forecasts that align closely with historical fluctuations.
- **“Storage”** exhibited gradual, steady growth and clear seasonal peaks mid-year, suggesting predictable demand cycles tied to inventory or business purchasing behavior.

- **“Machines”** displayed relatively stable but lower-amplitude seasonal movements, indicating steady demand with moderate forecast uncertainty.
- **“Binders”** showed irregular but generally stable sales, with modest increases expected through 2018; higher uncertainty toward the forecast horizon suggests sensitivity to market variation.
- Across all subcategories, forecast intervals (both 80% and 95%) expanded progressively over time, reflecting typical uncertainty in longer-term projections. Seasonal cycles remained evident, confirming a consistent 12-month pattern in category-level retail dynamics.

“Residual and Seasonality Observations:” Holt–Winters residuals were randomly distributed, showing a good model fit without major autocorrelation. Seasonal peaks were well captured, especially for **Phones** and **Storage**, while **Binders** and **Machines** showed weaker, less regular seasonality. **XGBoost** forecasts displayed higher variability during volatile months.

“Model Comparison and Interpretation:” Holt–Winters outperformed other models for most subcategories, offering accurate and interpretable results. The **Seasonal Naïve model** worked best for stable patterns like **Chairs**. **XGBoost** tended to **overfit** short-term fluctuations even after changing the **hyperparameters**. Overall, Holt–Winters provided the most reliable and practical forecasts for 2018 planning.

5. “Managerial Implications”

The forecasts provide clear guidance for aligning inventory, staffing, and marketing with demand cycles. Reliable monthly projections enable proactive planning and reduce operational uncertainty.

“Short-Term Actions (Quick Wins)”

- **Inventory Planning:** Build inventory buffers for **Phones** and **Chairs** ahead of peak months to prevent stockouts.

- **Promotion Timing:** Schedule marketing pushes during expected seasonal highs, such as mid-year peaks in **Storage** sales.
- **Staff Scheduling:** Adjust warehouse and delivery staffing to match forecasted surges, especially in high-volume categories.

“Long-Term Strategic Actions”

- **Product Focus:** Prioritize investment in **Phones** and **Storage**, which show consistent growth and strong profitability.
- **Portfolio Review:** Reevaluate **Binders** and **Machines**, where trends are flattening, consider redesign, bundling, or targeted promotions.
- **Capacity Planning:** Use 12-month forecasts to guide supplier orders and production scheduling, minimizing holding costs while meeting seasonal demand.

“**Managing Uncertainty:**” Forecast confidence intervals expand over longer horizons, highlighting increasing uncertainty. Managers should view forecasts as adaptable estimates rather than fixed targets. Implementing a rolling forecast process, updating models monthly with the latest sales data, will help capture shifts caused by promotions, market changes, or external factors.

“Monitoring Plan”

- Refresh data and retrain models **each month** to maintain relevance.
- Track forecast accuracy using **MAPE** and **RMSE**, adjusting parameters when errors exceed defined thresholds.
- Use **interactive dashboards** to monitor trends, seasonal patterns, and anomalies for real-time decision support.

These practices keep forecasts current, reliable, and aligned with evolving market dynamics.

“6. Reflection (Optional)”

The main challenges involved data inconsistencies, missing or duplicated records, irregular dates, and zero-profit months that interrupted continuity. Cleaning and aligning the dataset required more effort than the modeling itself. Another difficulty was balancing model complexity with interpretability: **XGBoost** captured nonlinear patterns but tended to **overfit** short sequences, while **Holt–Winters** provided smoother and more stable forecasts for seasonal data. Further tuning of XGBoost produced minimal accuracy gains at higher computational cost, showing that simpler models can perform competitively when data is well prepared.

Forecasting accuracy ultimately depended on the **entire pipeline**, not just the algorithm. Clean, consistent data and transparent evaluation metrics (MAE, RMSE, MAPE) across multiple time folds delivered the most reliable results. The project reinforced that careful data preparation and validation drive forecasting success.

“Quantitative Comparison of Excel and Python Results”

Python models achieved better accuracy than the Excel version. MAPE values were on average **2–5% lower**, with clear improvements for *Phones* (28.24% → 22.62%) and *Storage* (24.26% → 22.89%). MAE values dropped by roughly **20–30%**, with *Binders* improving the most—from \$16,104.75 to \$8,401.19 ($\approx 48\%$). The ranking pattern remained the same: *Phones*, *Chairs*, and *Storage* were most accurate, while *Binders* and *Machines* were less predictable. Overall, Python’s backtesting and model optimization reduced average errors and improved forecast reliability compared to Excel.